

PHYSICS

Q (1) deviate toward the normal

(2) IV

(3) Atmospheric refraction

(4) ciliary muscles

(5) 'A' is false and but 'R' is True

(6) $n = \frac{c}{v}$

$$n = 1.5$$

$$v = 2 \times 10^8 \text{ m/s}$$

$$nv = c$$

$$2 \times 10^8 \times 1.5 \times 10^8 = c$$

$$3 \times 10^8 \text{ m/s} = c$$

Q (8) $u = -25 \text{ cm}$

$$v = -75 \text{ cm}$$

We use convex lens

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{-75} + \frac{1}{25} = \frac{1}{f} = \frac{1}{-75} \quad f = 37.5 \text{ cm}$$

$$\rho = \frac{1}{f}$$

$$\rho = \frac{100}{f}$$

$$\rho = \frac{100}{37.5}$$

$$\rho = 2.66 \Omega$$

(g)

(i) Length = l
Resistance = R
Area = A

new Resistance where $2l$ on $A/2$

$$R_1 = \rho \frac{2l}{A/2}$$

$$R_1 = 4\rho \frac{l}{A}$$

$$R_1 = 4R$$

(ii) new resistivity

$$\cancel{R_1 = \rho}$$

$$4R =$$

(ii) Resistivity became unchanged

(10) (i) By applying maxwell's right hand thumb rule, the direction magnetic field is anti-clockwise the current.

(ii) At center of current carrying circular loop the direction of magnetic field is clockwise the current.

$\phi = 11$

(i) $I_1 = ?$, $R_1 = 10\Omega$

$I_2 = ?$, $R_2 = 15\Omega$

$V = 3V$

$$I_1 = \frac{V}{R_1}$$

$$I_1 = \frac{3}{10} = 0.3A$$

$$I_2 = \frac{V}{R_2}$$

$$I_2 = \frac{3}{15} = 0.2A$$

I_1 is greater than I_2

(ii) /

(iii) the ~~the~~ A and B are series combination but C is ~~parallel~~ parallel combination because A; B is smaller than B.

(12)

$$h_2 = +4 \text{ cm}$$

$$u = -25 \text{ cm}$$

$$f = -15 \text{ cm}$$

i)

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{f} - \frac{1}{u} = \frac{1}{v}$$

$$\frac{1}{-15} + \frac{1}{25} = \frac{1}{v}$$

$$\frac{-5 + 3}{75} = \frac{1}{v}$$

$$\frac{-2}{75} = \frac{1}{v}$$

$$v = -37.5 \text{ cm}$$

ii)

$$\frac{-v}{u} = \frac{h_1}{h_2}$$

$$\frac{+(-37.5)}{+25} = \frac{h_1}{4}$$

$$+1.5 = h_1$$

$$h_1 = +1.5 \text{ cm}$$

ii)

